A chatbot (large language model)

e.g. ChatGPT, Gemini, Claude

"This student has logged in 12 times and is 35. Will this student pass?"

↓

Trained on text
to predict the next word;
general-purpose by design

↓

"Students with low activity early in a course often struggle, so this student may be at some risk…"

reads sentences · returns a paragraph

A tabular foundation model

e.g. TabPFN, TabICL

	logins	age	credits	…	failed?
last year (outcome known)	34	22	60	⋯	0
	3	41	120	⋯	1
	18	29	60	⋯	0
	7	37	90	⋯	1
this year (to predict)	12	35	60	⋯	?
	2	24	120	⋯	?

↓

Trained on millions of synthetic tables
to fill in a missing column
from the rows above it

↓

risk of failing: **0.72** and **0.18**

reads numbers · returns a probability

What they share: both are *foundation models* — trained once, at large scale, then used without retraining. **What differs:** a chatbot answers questions; a tabular foundation model fills in a column of a spreadsheet.